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Misallocation, Selection, and Productivity: A Quantitative Analysis With Panel Data From China

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1 Big picture

- **Question.** How much do rural China’s land institutions lower agricultural productivity, and through which margins? The paper studies both the *static* margin—misallocation of land and capital across farmers—and the *selection* margin—which types of households remain in agriculture versus move to nonagriculture.
- **Core idea.** Weak and non-transferable land-use rights create wedges that fall most heavily on high-ability farmers. Those wedges not only prevent productive farmers from operating larger farms; they also distort occupational choice by reducing the effective return to farming for exactly the farmers who should be the most efficient operators.
- **Setting.** Rural China under the Household Responsibility System, with collective ownership of land, frequent village reallocations, thin rental markets, and very small farm size. The data cover 1993–2002.
- **Empirical strategy.** First, estimate permanent household fixed-effect measures of farm productivity and farm distortions from panel data, purging time shocks and village-level factors. Second, feed those moments into a two-sector Roy-style selection model with agriculture and nonagriculture.
- **Main within-agriculture result.** Efficient reallocation across existing farmers within villages raises agricultural TFP by 24.4%. Allowing reallocation across villages raises the gain to 53.2%.
- **Main selection result.** Eliminating the correlation between distortions and farm ability raises agricultural labor productivity by almost 3-fold, raises agricultural TFP by 67%, and lowers the agricultural employment share from 46% to 16%.
- **Amplification result.** Holding the set of farmers fixed, eliminating correlated distortions raises agricultural TFP by only 10%; once selection is allowed, agricultural TFP rises by 67%. Selection therefore roughly quadruples the productivity effect.
- **Why this paper matters.** The paper shows that distortionary institutions can lower productivity not only by misallocating factors *within* a sector, but also by worsening the composition of talent *across* sectors. That is a much broader mechanism than the standard static misallocation exercise.
- **Boundary of the analysis.** The paper does *not* estimate effects of land insecurity on long-run farm investment, irrigation, or technology adoption. The gains it measures therefore come from allocation and selection, so they are arguably conservative.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: why land policy should create wedges

- **Household Responsibility System.** Beginning in the late 1970s and early 1980s, rural households received use rights over farmland, but ownership remained with the collective.
- **Administrative allocation.** Village officials allocated land in an egalitarian way, largely based on household size rather than farming ability.

- **Frequent reallocations.** Even though the law promised secure use rights for 15 years, village officials often reallocated land earlier; in over two-thirds of villages reallocations occurred at least once, and on average more than twice.
- **Thin rental markets.** In 1995 households rented out less than 3% of their land, despite many villages formally reporting no restrictions on rental activity.
- **“Use it or lose it.”** Renting out land or leaving it idle could be interpreted by village officials as evidence the household did not need the land, so households feared losing their use rights in the next reallocation.
- **Capital-market implication.** Because households did not own the land, they could not use it as collateral, so land institutions plausibly distorted capital allocation as well.
- **Scale implication.** Average farm size in China was only about 0.7 hectares, far below developed-country benchmarks, and did not rise over time.

2.2 Household panel data

- **Source.** Household survey data collected by the Research Center for the Rural Economy under China’s Ministry of Agriculture.
- **Coverage.** Ten provinces spanning the major regions of China, over 1993–2002.
- **Panel structure.** Roughly 8,000 households per year from 110 villages; 104 villages are observed for all nine years; about 6,000 households appear in all nine years.
- **Village-year density.** The average village-year contains about 80 household observations, or roughly one-quarter to one-third of village households.
- **Attrition.** The paper argues attrition is not a major concern; much of it comes from entire villages leaving the survey rather than selective household exit.

2.3 Measured farm and nonfarm objects

- **Agricultural variables.** Household land holdings, sown area, crop outputs, labor, fertilizer, and farm machinery.
- **Nonagricultural variables.** Wage earnings plus revenues, expenditures, and net incomes from household businesses.
- **Real farm output.** The authors use common sample-wide average prices for crops and intermediate inputs over 1993–2002 to construct real value added. This avoids confounding farm productivity with farm-specific price variation.
- **Normalization by labor days.** The theoretical framework abstracts from within-household differences in the intensive margin of labor. To map the data into the model, value added, land, and capital are normalized by total agricultural labor days.

2.4 Parameterization used to back out farm productivity

- **Returns to scale.** $\gamma = 0.54$, so the labor share is $1 - \gamma = 0.46$.

- **Land versus capital.** $\alpha = 2/3$, implying a land share of 0.36 and a capital share of 0.18.
- **Interpretation.** Most of the residual variation in farm value added, after controlling for normalized land and capital, is interpreted as farm-level physical productivity.

2.5 Permanent productivity and distortion measures

- **Permanent farm productivity.** The baseline TFP_i is the household fixed effect from panel regressions, after removing year effects and then purging village fixed effects.
- **Permanent distortions.** The baseline $TFPR_i$ is constructed the same way from revenue productivity.
- **Why this matters.** Relative to cross-sections, this design strips out transitory shocks, measurement noise, and village-level heterogeneity.
- **Dispersion reduction.** The standard deviation of log farm TFP falls from about 0.72 in the average cross-section to 0.35 in the baseline fixed-effect measure.
- **Land-quality check.** Using FAO GAEZ data, village-level land-quality dispersion has standard deviation 0.096 in logs, which the authors interpret as an upper bound on within-village land-quality heterogeneity. They argue that within-village land quality is unlikely to explain the much larger dispersion in household TFP.

3 Models and empirical specifications (all equations + interpretation)

3.1 Farm production function

For farmer i in village v at time t , output is

$$y_{ivt} = (A_a s_{ivt})^{1-\gamma} \ell_{ivt}^{\alpha\gamma} k_{ivt}^{(1-\alpha)\gamma}. \quad (1)$$

- **Interpretation.** A farm combines operator ability s_{ivt} , land ℓ_{ivt} , and capital k_{ivt} . The exponent $\gamma < 1$ creates decreasing returns at the farm level, so more able farmers should operate larger farms but not infinitely large ones.
- **Economic role of ability.** Ability enters as a Hicks-neutral shifter. More able farmers have higher output for any given input bundle.
- **Why decreasing returns matter.** They pin down finite optimal farm size and make the efficient allocation problem well behaved.

3.2 Efficient allocation benchmark implied by the planner's problem

Given aggregate village land L_{vt} and capital K_{vt} , the efficient allocation assigns factors in proportion to farm ability:

$$\ell_{ivt}^e = \frac{s_{ivt}}{\sum_{j=1}^{M_{vt}} s_{jvt}} L_{vt}, \quad k_{ivt}^e = \frac{s_{ivt}}{\sum_{j=1}^{M_{vt}} s_{jvt}} K_{vt}. \quad (2)$$

Aggregate efficient output is then

$$Y_{vt}^e = A_a^{1-\gamma} L_{vt}^{\alpha\gamma} K_{vt}^{(1-\alpha)\gamma} \left(\sum_{i=1}^{M_{vt}} s_{ivt} \right)^{1-\gamma}. \quad (3)$$

- **Interpretation.** In the efficient benchmark, more able farmers should control proportionally more land and capital.
- **Empirical implication.** If actual Chinese land institutions are efficient, land and capital should be strongly positively related to farm ability, and marginal products should be flat with respect to ability.
- **What the paper tests.** The data show the opposite: scale barely rises with productivity, while average products do.

3.3 Revenue productivity as a summary measure of farm-specific distortions

$$\text{TFPR}_{ivt} = \frac{y_{ivt}}{\ell_{ivt}^\alpha k_{ivt}^{1-\alpha}}. \quad (4)$$

- **Interpretation.** TFPR is the revenue-productivity object familiar from Hsieh–Klenow style exercises. Here it summarizes the combined effect of land and capital wedges relative to any output wedge.
- **Economic meaning.** In an undistorted allocation, TFPR should be equalized across farms. Dispersion in TFPR indicates distortions.
- **Key paper insight.** The paper does not just find dispersion in TFPR; it finds that TFPR is *positively correlated* with farm productivity, which means the more able farmers are precisely the more distorted ones.

3.4 Physical productivity (farm TFP)

$$\text{TFP}_{ivt} \equiv (A_a s_{ivt})^{1-\gamma} = \frac{y_{ivt}}{\left(\ell_{ivt}^\alpha k_{ivt}^{1-\alpha} \right)^\gamma}. \quad (5)$$

- **Interpretation.** TFP is the physical productivity object. It is the farm's output residual after netting out input quantities with the assumed production elasticities.
- **Contrast with TFPR.** TFP captures how productive the farmer is; TFPR captures how distorted the farmer's allocation is.
- **Undistorted benchmark.** High-TFP farmers should operate more land and capital, while TFPR should be flat.

3.5 Panel fixed-effect measure of permanent farm productivity

The first-stage panel decomposition is

$$\log \text{TFP}_{it} = \mu_t^{\text{TFP}} + \mu_i^{\text{TFP}} + e_{it}^{\text{TFP}}, \quad (6)$$

followed by

$$\mu_i^{\text{TFP}} = \mu_v^{\text{TFP}} + \zeta_i^{\text{TFP}}. \quad (7)$$

- **Interpretation.** The first regression removes year shocks; the second strips village-specific time-invariant factors from the household fixed effect.
- **Baseline object.** The empirical TFP_i is $\exp(\hat{\zeta}_i^{\text{TFP}})$.
- **Why this matters.** This is the core reason the paper claims to improve on cross-sectional misallocation studies. It isolates the permanent, farmer-specific component of productivity.

3.6 Panel fixed-effect measure of permanent distortions

The same decomposition is applied to revenue productivity:

$$\log \text{TFPR}_{it} = \mu_t^{\text{TFPR}} + \mu_i^{\text{TFPR}} + e_{it}^{\text{TFPR}}, \quad (8)$$

and then

$$\mu_i^{\text{TFPR}} = \mu_v^{\text{TFPR}} + \zeta_i^{\text{TFPR}}. \quad (9)$$

- **Baseline object.** Permanent distortions are summarized by $\text{TFPR}_i = \exp(\hat{\zeta}_i^{\text{TFPR}})$.
- **Key empirical result.** The correlation between $\log \text{TFP}_i$ and $\log \text{TFPR}_i$ is 0.91. This is the central reduced-form fact of the paper.
- **Interpretation.** High-ability farmers are the ones facing the largest wedges, exactly as the institutional story would predict.

3.7 Mismeasurement test using panel responses of output to inputs

The paper applies the Bils–Klenow–Ruane approach and estimates

$$\Delta \log(y_{it}) = \beta_1 \log(\text{TFPR}_{it}) + \beta_2 \Delta \log(I_{it}) + \beta_3 [\Delta \log(I_{it}) \times \log(\text{TFPR}_{it})] + \mu_v + \mu_t + u_{it}, \quad (10)$$

where the input bundle is

$$I_{it} = \ell_{it}^\alpha k_{it}^{1-\alpha}, \quad \hat{\lambda} = 1 + \frac{\hat{\beta}_3}{\hat{\beta}_2}. \quad (11)$$

- **Interpretation.** $\hat{\lambda}$ measures the share of TFPR dispersion due to true distortions rather than additive measurement error. Values close to one mean little mismeasurement.
- **Quantitative finding.** With the baseline fixed-effect TFPR measure, $\hat{\lambda} = 1.00$, versus 0.96 for household-plus-village fixed effects and 0.90 in raw cross-sections.
- **Takeaway.** The paper’s main results are not being driven by the type of additive noise that often plagues cross-sectional misallocation estimates.

3.8 Efficiency-gain statistic

The productivity gain from eliminating distortions is measured as

$$\text{Gain} = \frac{Y^e}{Y} - 1. \quad (12)$$

- **Interpretation.** Aggregate land, capital, and number of farmers are held fixed, so the output gain is also a TFP gain.
- **Benchmark exercise.** The paper compares the actual Chinese allocation to the efficient allocation implied by the planner problem.

3.9 Two-sector selection model: sectoral incomes

An individual working in nonagriculture earns

$$I_i^n = w_n s_{ni}, \quad (13)$$

while an individual operating a farm earns

$$I_i^a = w_a \phi_i s_{ai}. \quad (14)$$

- **Interpretation.** In agriculture, ability s_{ai} is multiplied by a farmer-specific distortion ϕ_i . A lower ϕ_i means a larger wedge and therefore a lower effective agricultural return.
- **Key conceptual novelty.** Distortions affect occupational choice *directly*, not only through general-equilibrium prices.

3.10 Occupational choice condition

A household chooses agriculture if

$$w_a \phi_i s_{ai} \geq w_n s_{ni}. \quad (15)$$

- **Interpretation.** What matters is effective agricultural ability $\phi_i s_{ai}$, not just raw agricultural ability s_{ai} .
- **Amplification mechanism.** If high-ability farmers are also the ones with low ϕ_i , they can be pushed out of agriculture, reducing both mean ability among remaining farmers and aggregate agricultural productivity.

3.11 Calibration discipline

- **Joint distribution.** The model assumes a trivariate log-normal distribution for (s_{ai}, ϕ_i, s_{ni}) .
- **Micro moments used.** The calibration uses conditional moments of agricultural income, nonagricultural income, and distortions based on the panel fixed-effect measures.
- **Especially important moment.** About 96% of households have income observed in both sectors, so the paper can use the contemporaneous correlation of log sectoral incomes across households, 0.034, to discipline the joint distribution of abilities.
- **Aggregate target.** The benchmark economy is calibrated to match the 46% agricultural employment share in China.
- **Model fit.** The model reproduces the data's gain from eliminating correlated distortions across existing farmers reasonably well: 10% in the model versus 13.9% in the data.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea: no external IV, but strong structural discipline

- **Core comparison.** The paper compares actual input allocations to the efficient benchmark implied by the production function and a planner problem with the same aggregate land, capital, and farmers.
- **Why within-village variation matters.** The preferred results purge village effects, so the wedges are identified from differences *across households inside the same institutional and local environment*, not from differences in prices or land quality across villages.
- **Main structural premise.** Under egalitarian land allocation and thin rental markets, ability is not allowed to map into scale as it would under secure, transferable rights.
- **What would refute the story?** If productive farmers already operated systematically larger farms and marginal products were flat, the institutional interpretation would collapse. Figure 1 shows that this is not the case.

4.2 Why the paper argues the wedges are not just measurement error

- **Panel fixed effects.** The use of household fixed effects removes transitory farm-specific shocks and year effects.
- **Village purging.** Regressing household fixed effects on village dummies removes time-invariant village heterogeneity, including common local conditions.
- **BKR diagnostic.** The additive measurement-error estimate $\hat{\lambda} = 1.00$ is very high for the baseline TFPR measure.
- **Land-quality argument.** Village-level land-quality dispersion from GAEZ is small relative to household TFP dispersion, and the institutional detail about bundled land allocations further weakens the land-quality alternative.

- **Caveat.** The BKR test only detects certain forms of additive measurement error. The authors are careful not to claim that every form of mismeasurement has been eliminated.

4.3 Evidence tying the wedges specifically to land institutions

- **Institutional match.** The fact that the most productive farmers face the largest distortions is exactly what one would expect under nearly egalitarian land allocation with limited ability to rent in land.
- **Developed-country comparison.** In U.S. farm data, farm size is strongly positively correlated with labor productivity and alternative TFP measures, unlike in China.
- **Rental-market comparison within China.** In provinces with relatively more land rental activity, the correlation between land input and farm productivity rises from 0.02 in the full sample to 0.13. That is still small, but it points in the direction predicted by the mechanism.
- **Interpretation of TFPR.** Because land institutions likely distort both land and capital, the paper interprets a large part of TFPR variation as an output wedge associated with land policy, while the residual capital-to-land wedge captures other factors.

4.4 Identification of the selection channel

- **Key identification advantage.** The survey records both farm and nonfarm incomes for most households, so the model is not forced to infer the unchosen sector's income from thin assumptions alone.
- **What disciplines selection.** The observed cross-sector income correlation, the large observed movement out of agriculture, and the agricultural employment share together pin down how distortions and abilities interact.
- **Why selection is separate from misallocation.** The paper compares the full equilibrium effect of eliminating correlated distortions to the partial-equilibrium effect of holding the set of farmers fixed. The gap between 1.67 and 1.10 for agricultural TFP isolates the amplification from selection.

5 Tables and figures

5.1 Figure 1: Factor allocations by farm productivity

Read:

- The figure has four panels: land vs. farm productivity, capital vs. farm productivity, land productivity vs. farm productivity, and capital productivity vs. farm productivity.
- In an efficient allocation, the dashed benchmark says land and capital should rise strongly with farm productivity, while average products should be flat.
- In the data, land is essentially flat in farm productivity and capital is even slightly negatively related to farm productivity. Meanwhile, average land and capital productivity rise sharply with farm productivity.

- This is the cleanest visual summary of misallocation in the paper: the best farmers are not the ones operating the largest farms, but they do have the highest marginal products.

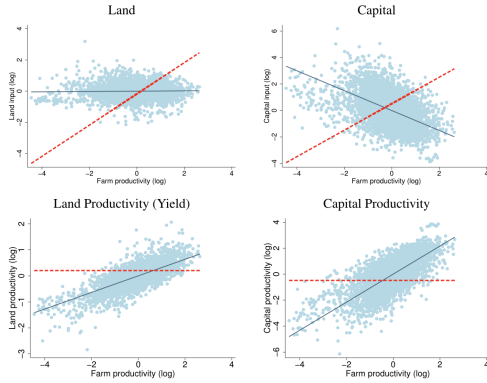


FIGURE 1.—Factor Allocations by Farm Productivity. Notes: The data on inputs and productivities refer to the permanent household fixed-effect measures purged of time and village-level factors. Land and capital are measured relative to total labor days supplied to agriculture by the household. The solid line is the estimated relationship between inputs and farm productivity whereas the dashed line is the efficient allocation associated with each level of farm productivity. Land productivity refers to value added per unit of land and capital productivity refers to value added per unit of capital, both of which are proportional to the marginal products of each factor in our framework.

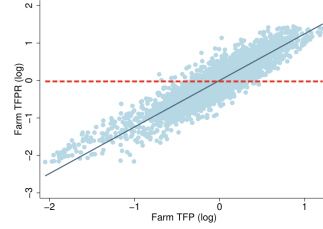


FIGURE 2.—Farm-Specific Distortions and Productivity. Notes: $TFPR_i$ is a summary measure of distortions faced by each household. The standard deviation of log $TFPR_i$ is 0.48 and the correlation between log TFP_{*i*} and log $TFPR_i$ is 0.91.

5.2 Figure 2: Farm-specific distortions and productivity

Read:

- This figure plots the baseline permanent $TFPR_i$ measure against permanent TFP_i .
- The relationship is strongly positive, with correlation 0.91 and standard deviation of log $TFPR_i$ equal to 0.48.
- Economically, this means the more productive the farmer, the larger the distortion the farmer faces. That pattern is exactly why selection effects become so large in the model.

5.3 Table I: Mismeasurement in productivity and distortions

Read:

- The table compares three measurement strategies: household fixed effects purged of village effects, household-plus-village fixed effects, and raw cross-sections.
- Dispersion in both TFP and TFPR falls sharply as the paper moves from cross-sections to panel fixed effects. For example, the standard deviation of log TFP falls from 0.72 to 0.35.
- The BKR statistic rises to 1.00 for the baseline measure, implying essentially no detectable additive measurement error of the relevant type.
- The correlation between log TFP and log TFPR remains strong even after the noise is stripped out, which supports the idea that the paper's main fact is structural rather than spurious.

5.4 Table II: Efficiency gains from reallocation

Read:

TABLE I
MISMEASUREMENT IN PRODUCTIVITY AND DISTORTIONS.

	Fixed-Effect Estimates		Cross-Section Average
	Household Farm	+ Village	
Farm TFP:			
STD(log)	0.35	0.64	0.72
p90/p10	2.19	4.35	5.59
p75/p25	1.48	2.06	2.32
Farm TFPR:			
STD(log)	0.48	0.81	0.92
p90/p10	3.14	7.17	9.70
p75/p25	1.78	2.71	3.23
CORR (logTFP, logTFPR)	0.91	0.88	0.88
BKR $\hat{\lambda}$	1.00	0.96	0.90
Standard error	(0.026)	(0.039)	(0.024)
95% confidence interval	[0.95, 1.05]	[0.88, 1.03]	[0.85, 0.95]

Note: The first two columns refer to statistics for fixed-effect estimates of TFP and TFPR from panel regressions. The column "Household Farm" refers to the fixed-effects estimates, after removing village specific factors. The column "+ Village" refers to the fixed-effect estimates, inclusive of village-level factors but excluding land quality differences across villages. "Cross-section average" refers to statistics computed in each year and then averaged across years. "BKR $\hat{\lambda}$ " is the estimate of the fraction of TFPR variation that is not measurement error as in Bils, Klenow, and Ruane (2017). Each BKR $\hat{\lambda}$ is based on an OLS regression of changes in log-output on changes in log-inputs, an interaction of changes in log-inputs and TFPR, with village and year dummies, and clustering of standard errors at the village level. Standard errors are in parentheses and 95% confidence intervals in square brackets.

TABLE II
EFFICIENCY GAINS FROM REALLOCATION.

	Output (TFP) Gain (%)			Cross-Section Average
	Total	Across s Misallocation	Land Distortion	
Eliminating misallocation across households:				
within villages	24.4	13.9	13.6	54.0
+ across villages	53.2	24.9	–	83.0

Note: Output (TFP) gain from efficient reallocation in percent. "Total" refers to the fixed-effects estimates from the panel regression. "Across s misallocation" refers to the reallocation gains only eliminating misallocation across farm households with different productivity. "Land distortion" refers to the efficiency gain arising from eliminating the output wedge arising from the land institution. "Cross-section average" refers to reallocation gains when farm TFP and distortions are computed for each cross-section and reallocation gains calculated for each year in the panel and then averaged across years.

- Within villages, eliminating misallocation raises agricultural output and TFP by 24.4%.
- Of that, 13.9% comes from reallocation across farmers with different productivity levels—the correlated-distortion component that matters most for the later selection exercise.
- Interpreting the land institution as an output wedge yields a 13.6% gain, or almost 60% of the within-village total.
- Allowing reallocation across villages raises the total gain to 53.2%, more than doubling the within-village number.
- Cross-sectional methods would imply much larger gains, 54% within villages and 83% across villages, which the paper takes as evidence that traditional cross-sectional exercises overstate reallocation gains.

5.5 Table III: The effects of correlated distortions

Read:

- The table compares the benchmark economy to a counterfactual that removes only the component of farm distortions correlated with agricultural ability.
- Agricultural labor productivity rises from 1.00 to 2.96, agricultural TFP rises from 1.00 to 1.67, and the agricultural employment share falls from 0.46 to 0.16.
- Holding the benchmark set of farmers fixed, agricultural TFP would rise only to 1.10. The much larger rise to 1.67 therefore comes from better selection into agriculture.
- Real GDP per worker rises only 18%, reminding us that large agricultural productivity gains need not translate one-for-one into aggregate GDP gains when labor shifts into another sector whose average productivity then falls.
- The conditional correlation of farm TFP and farm TFPR falls from 0.97 to 0.44, and the cross-sector income correlation among agricultural choosers rises from 0.03 to 0.40, which is exactly what one would expect when high-ability farmers are no longer disproportionately taxed by the land system.

TABLE III
THE EFFECTS OF CORRELATED DISTORTIONS.

Statistic	Benchmark Economy (BE)	No Correlated Distortions
Aggregate statistics		
Real agricultural productivity (Y_a/N_a)	1.00	2.96
Share of employment in agriculture (N_a) (%)	0.46	0.16
TFP in agriculture (TFP _a)	1.00	1.67
TFP in agriculture, constant BE farms	1.00	1.10
Real nonagricultural productivity (Y_n/N_n)	1.00	0.78
Average ability in agriculture (Z_a/N_a)	1.00	2.34
Average ability in nonagriculture (Z_n/N_n)	1.00	0.78
Real GDP per worker (Y/N)	1.00	1.18
Conditional microlevel statistics		
STD of log-farm TFP	0.56	0.39
STD of log-farm TFPR	0.48	0.14
CORR of log-(farm TFP, farm TFPR)	0.97	0.44
CORR of log-(agr. ability, nonagr. ability)	0.15	0.49
CORR of log-(agr. income, nonagr. income)	0.03	0.40

Note: “No Correlated Distortions” eliminates the component of farm-level distortions ϵ_i associated with agricultural ability. Aggregate variables, except for the share of employment in agriculture, reported relative to the same statistic in the Benchmark Economy (BE). Micro-level statistics reported in levels, and are conditional on choosing agriculture in the corresponding simulated economy. Real agricultural productivity is agricultural output divided by employment in agriculture. TFP_a is agricultural output divided by the bundle of inputs from the implied aggregate production function.

6 Short summary

- **Conclusion.** Rural China’s land institutions create large wedges that keep productive farmers from expanding and also push some of them out of agriculture entirely.
- **Intuition.** Equalizing access to land may look egalitarian, but when land cannot move to the best farmers, the economy loses twice: resources are badly allocated among existing farmers, and the wrong people remain in farming.
- **Empirical lesson.** Panel fixed effects materially improve misallocation measurement. Cross-sectional estimates exaggerate both dispersion and reallocation gains.
- **Quantitative lesson.** Selection is not a second-order add-on. In this paper it is the dominant channel through which land institutions depress agricultural productivity.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Capital and land are severely misallocated across farmers within villages in China, and these distortions are systematically larger for more productive farmers.
- **Magnitude.** Within-village reallocation raises agricultural TFP by 24.4%, while broader across-village reallocation raises it by 53.2%.
- **Selection amplification.** Eliminating only the correlation between distortions and farm ability raises agricultural TFP by 67%, far above the 10% gain obtained when the set of farmers is held fixed.
- **Measurement contribution.** By using panel fixed effects and the BKR diagnostic, the paper shows that the key dispersion patterns are not mainly an artifact of additive measurement error.
- **Policy implication.** Secure and transferable land-use rights would let land flow toward high-ability farmers, improve farm size, keep better farmers in agriculture, and speed structural transformation.

7.2 Economic intuition

- A good farmer should want more land and complementary capital. Egalitarian land assignment blocks that expansion.
- Because land cannot be used freely or pledged as collateral, the distortion affects both land scale and access to capital.
- The farmers hurt most are the highest-ability ones, since they stand to gain the most from expansion. That is why the wedge is positively correlated with productivity.
- Once those farmers face lower effective agricultural returns, some leave farming. Agriculture then loses not only factor efficiency but also talent.

7.3 Takeaway

This paper is a very strong example of how to connect an observed institution to a misallocation object and then carry that object into a general-equilibrium model of selection. The static misallocation result is already substantial, but the real contribution is showing that wedges aimed at productive farmers change who farms at all. In the paper's accounting, that selection margin is what turns a meaningful productivity loss into a very large one.